Looking Beyond the Window: Global-Local Aligned CLIP for Training-free Open-Vocabulary Semantic Segmentation

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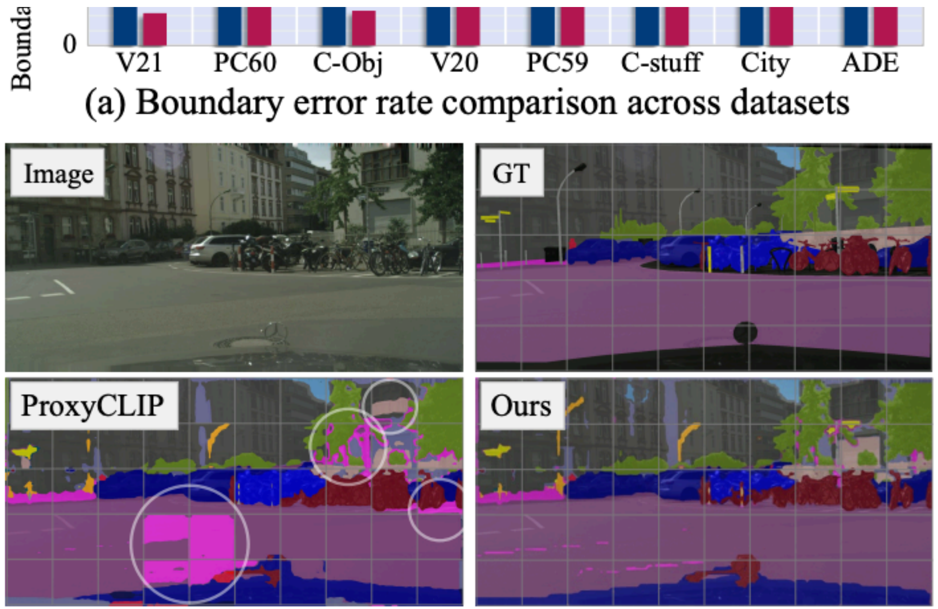
できた: 課題を言い切った (ref: 3.1.2)

原因は学習不足ではなく、window ごとの文脈分断

高解像度 OVSS では sliding-window 推論が必要だが、window を独立に処理すると同じ物体でも予測がずれる。

- 境界で grid artifact が出る
- large object と stuff category で特に悪化
- BER でその不整合を定量化

重なり領域を平均しても、意味のズレ自体は直らない。



(b) Grid artifacts from contextual discrepancy across windows

Figure 1. Comparison of segmentation consistency near window boundaries. (a) We evaluate segmentation inconsistency using the Boundary Error Rate (BER). BER is defined as the proportion of pixels near adjacent window boundaries where predicted labels differ despite identical ground-truth. ProxyCLIP yields high BER due to its lack of cross-window interaction, whereas BER is significantly reduced in ours by incorporating global context into attention process. More details in Appendix A. (b) ProxyCLIP exhibits

Figure 1: Window 境界の不整合が主要な失敗

できた: 推論時だけで補正した (ref: 3)

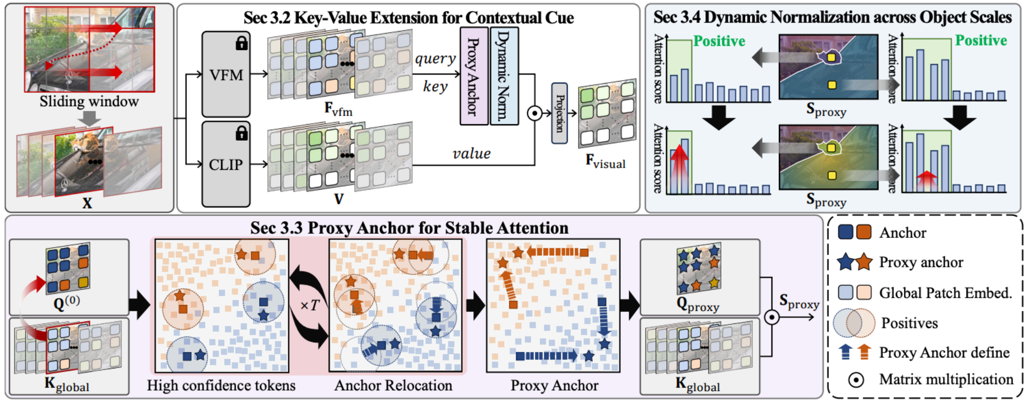


Figure 2. Overview of our proposed framework. The input image is first divided into overlapping windows and processed using frozen backbones: a Vision Foundation Model (VFM, e.g., DINO) and CLIP. We introduce a Key-Value Token Extension, where VFM features from the current window $\mathbf{F}_{\text{vfm}}$ serve as *query* tokens, while *key* tokens are gathered from all windows to provide global context. The corresponding *value* tokens V are extracted from the final transformer layer of CLIP. Cross-attention is then applied, followed by a projection layer to generate the final visual features $\mathbf{F}_{\text{visual}}$. To stabilize attention across windows, each query token is replaced with a semantically representative proxy anchor. Finally, a dynamic normalization scheme adjusts attention strength based on object size, approximated by the number of positive samples associated with each proxy anchor.

Figure 2: GLA-CLIP の全体像。

再学習なしで、window をまたぐ受容野を回復

- **Key-Value Extension**: 参照先だけ全 window に広げる
- **Proxy Anchor**: local query の window bias を弱める
- **Dynamic Normalization**: object scale ごとに global attention を調整

単に global context を足すだけでなく、bias と scale のずれも同時に補正する。

できた: 結果が出た (ref: 4.2)

既存法に後付けで効き、

window artifact も減る

- 8 benchmark で一貫して改善
- CLIP-DINOiser / SCLIP では平均 **+1.6 mIoU**
- tuning なしで **44.0 mIoU**

性能改善が「表の数値」だけでなく「見た目の整合性」にも出ている点が強い。

Model	Train	External Model	*DS Hyperparameter	With Background V21	PC60	C-Obj	V20	Without Background PC59	C-Stf	City	ADE	Avg
CLIP-DINOiser Setting	448 Image Resize, 448 Crop size, 224 Stride, Without rename trick†											
GroupViT [46]	✓	-	✘	50.4	18.7	27.5	79.7	23.4	15.3	11.1	9.2	29.4
ReCo [40]	✓	-	✘	25.1	19.9	15.7	57.7	22.3	14.8	21.6	11.2	23.5
CLIP-DIY [44]	✘	-	✘	59.9	19.7	31.0	79.7	19.8	13.3	11.6	9.9	30.6
TCL [7]	✓	-	✘	55.0	30.4	31.6	**83.2**	33.9	22.4	24.0	17.1	37.2
CLIP-DINOiser [45]	✓	DINO	✘	62.2	32.4	35.0	80.2	35.9	24.6	31.7	**20.0**	<u>40.3</u>
ProxyCLIP + GLA‡	✘	DINO	✘	**63.4**	**35.8**	**37.3**	<u>80.4</u>	**39.4**	**26.3**	**33.5**	<u>19.1</u>	**41.9**
ClearCLIP Setting	448 Image Resize, 448 Crop size, 224 Stride, Without rename trick											
ClearCLIP [23]	✘	-	✘	51.8	32.6	33.0	**80.9**	35.9	23.9	30.0	16.7	<u>38.1</u>
ClearCLIP + GLA	✘	-	✘	**55.6**	**33.2**	**34.6**	<u>79.7</u>	**36.5**	**24.6**	**32.7**	**17.4**	**39.3**
ProxyCLIP Setting	448 Image Resize, 336 Crop size, 112 Stride, With rename trick on Background Classes											
ProxyCLIP [24]	✘	DINO	✘	61.3	35.3	37.5	80.3	39.1	26.5	38.1	20.2	<u>42.3</u>
ProxyCLIP + GLA	✘	DINO	✘	**63.2**	**35.8**	**37.7**	**81.5**	**39.7**	**26.8**	**38.5**	**20.3**	**42.9**
SCLIP Setting	336 Image Resize, 224 Crop size, 112 Stride, With rename trick											
SCLIP [42]	✘	-	✘	59.1	30.4	30.5	80.4	34.2	22.4	32.2	16.1	<u>38.2</u>
SCLIP + GLA	✘	-	✘	**59.8**	**32.0**	**32.8**	**80.5**	**35.1**	**24.2**	**35.1**	**18.0**	**39.8**
LaVG [20]	✘	-	✘	62.1	31.6	34.2	82.5	34.7	23.2	26.2	15.8	38.8
NACLIP [18]	✘	-	✘	58.9	32.2	33.2	79.7	35.2	23.3	35.5	17.4	39.4
ResCLIP [50]	✘	-	✓	61.1	33.5	35.0	<u>86.0</u>	36.8	24.7	35.9	18.0	41.4
FreeCP [8]	✘	-	✓	65.8	35.3	37.2	84.3	38.0	24.9	33.3	18.4	42.2
DIH-CLIP [16]	✘	-	✘	64.2	36.0	37.4	84.9	39.7	24.5	40.2	19.6	43.3
FLOSS [2]	✘	-	✘	-	-	-	80.2	35.9	23.6	37.0	18.4	-
CASS [22]	✘	DINO	✓	65.8	**36.7**	**37.8**	**87.8**	**40.2**	26.7	39.4	<u>20.4</u>	**44.4**
ProxyCLIP + GLA (Best§)	✘	DINO	✓	**66.7**	<u>36.3</u>	<u>37.7</u>	84.7	**40.2**	**27.2**	**41.2**	**20.5**	44.3
ProxyCLIP + GLA	✘	DINO	✘	<u>66.3</u>	36.1	<u>37.7</u>	84.2	<u>39.9</u>	<u>26.9</u>	<u>40.8</u>	20.0	<u>44.0</u>

Table 1. Open-vocabulary semantic segmentation results on 8 datasets using CLIP ViT-B/16. *DS Hyperparameter: dataset-specific hyperparameters (not used in Ours). †Rename trick: multiple textual prompts per class. ‡GLA: Adaptation for ClearCLIP [23], SCLIP [42], and ProxyCLIP [24]. § Best: optionally uses dataset-specific hyperparameters (**u**, **w**), by manual tuning.

Figure 3: 複数手法への追加で安定した利得。

できた: 見た目でも一貫した (ref: 4.2)

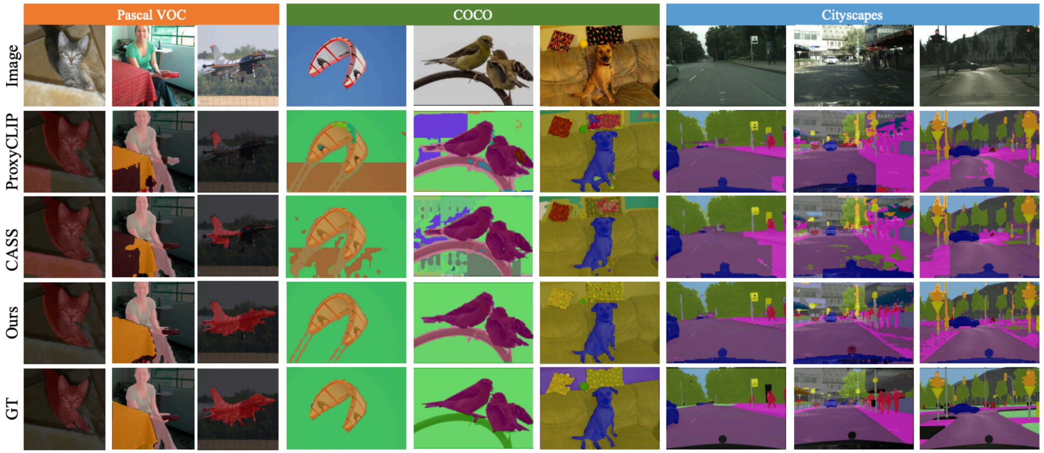


Figure 4. Qualitative results among ProxyCLIP [24], CASS [22], Ours on Pascal VOC21 [17], COCOstuff [4], and Cityscapes [12].

Figure 4: Window をまたぐ予測の段差が小さい。

同じ領域に、同じ意味を返しやすくなった

- Pascal VOC / COCO-Stuff / Cityscapes で改善
- 背景や大きい物体のラベルが安定
- BER 改善が視覚的一貫性に対応

この論文の説得力は、artifact を定量・定性の両方で押さえている点にある。

できない (ref: 6)

全 window の token を見るので、推論は重い

- 計算量とメモリコストが増える
- 高品質 token を選別する仕組みまでは入っていない
- 学習なしで効く代わりに、効率面はまだ未解決

今後 (ref: 6)

次は「全部見る」から「必要な token だけ見る」へ

- 疎な global retrieval で計算量を落とす
- object scale に応じた token selection を明示化する
- training-free のまま、より大きい画像と複雑な scene に広げる